

# Complements of Operational Research

8.1 Multi-criteria decision making: the AHP, pairwise comparisons and consistency

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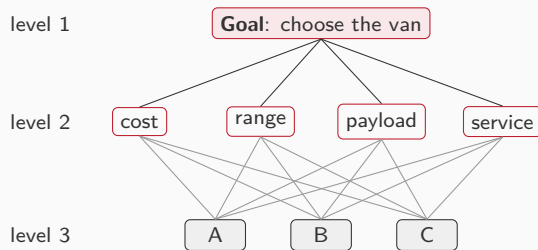
# When there is no common unit of measure

In chapters 2 to 7 there was always **one** measure: time, cost, number of customers waiting. Many real decisions are not like that.

An urban delivery company is buying electric vans and hesitates between three models, A, B and C. What matters:

- the purchase **cost** (€);
- the **range** (km);
- the **payload** (kg);
- the after-sales **service** (no unit: an opinion).

Nobody knows how many euros one kilometre of range is worth.



**AHP** (*Analytic Hierarchy Process*, Saaty, 1970s):  
break the decision into a hierarchy, compare **in pairs** at each level and combine everything into one number per alternative.

# Saaty's scale

| $a_{ij}$   | $i$ compared with $j$        |
|------------|------------------------------|
| 1          | equal importance             |
| 3          | moderately more important    |
| 5          | strongly more important      |
| 7          | very strongly more important |
| 9          | extremely more important     |
| 2, 4, 6, 8 | intermediate values          |

**Reciprocity:**  $a_{ji} = 1/a_{ij}$  and  $a_{ii} = 1$ . With  $n$  criteria only  $n(n-1)/2$  judgements are needed (here 6).

**How it appears in the tests:** one ruler per pair of criteria.



A mark on the left favours the criterion on the left. Here the mark is on the 3 to the right: range is moderately more important than cost, that is,  $a_{\text{cost}, \text{range}} = 1/3$ .

A ratio scale: “5” means “five times more important”.

# The criteria weights

The company's management compared the four criteria:

|         | cost  | range | payl. | serv. |
|---------|-------|-------|-------|-------|
| cost    | 1     | 2     | 3     | 5     |
| range   | 1/2   | 1     | 2     | 3     |
| payload | 1/3   | 1/2   | 1     | 2     |
| service | 1/5   | 1/3   | 1/2   | 1     |
| sum     | 2.033 | 3.833 | 6.5   | 11    |

**Step 1:** divide each column by its sum.

**Step 2:** the weight of each criterion is the average of its row.

|         | normalised matrix |       |       |       | weight $w$   |
|---------|-------------------|-------|-------|-------|--------------|
| cost    | 0.492             | 0.522 | 0.462 | 0.455 | <b>0.482</b> |
| range   | 0.246             | 0.261 | 0.308 | 0.273 | 0.272        |
| payload | 0.164             | 0.130 | 0.154 | 0.182 | 0.158        |
| service | 0.098             | 0.087 | 0.077 | 0.091 | 0.088        |

Cost accounts for almost half of the decision; service for less than 9%.

If the judgements were perfectly consistent ( $a_{ij} = w_i/w_j$ ), the four normalised columns would be equal: the average only “cleans up” small differences.

# Are the judgements consistent?

If cost = 2× range and range = 2× payload,  
consistency requires cost ≈ 4× payload.  
Management said 3: close.

## Consistency index and ratio

$$\lambda_{\max} \approx \frac{1}{n} \sum_i \frac{(Aw)_i}{w_i}, \quad CI = \frac{\lambda_{\max} - n}{n - 1}, \quad CR = \frac{CI}{RI_n}$$

The matrix is accepted if  $CR < 0.10$ .

$\lambda_{\max} \geq n$ , with equality only if the matrix is perfectly consistent.

**Vans:**  $Aw = (1.940, 1.093, 0.631, 0.354)$   
 $(Aw)_i/w_i = 4.021, 4.021, 4.005, 4.011$   
 $\lambda_{\max} = 4.015$ ;  $CI = 0.0048$ ;  
 $CR = 0.0048/0.90 = \mathbf{0.005}$ : consistent.

Saaty's random index

| $n$    | 3    | 4    | 5    | 6    | 7    | 8    | 9    |
|--------|------|------|------|------|------|------|------|
| $RI_n$ | 0.58 | 0.90 | 1.12 | 1.24 | 1.32 | 1.41 | 1.45 |

$RI_n$ : average CI of randomly filled matrices. For  $n \leq 2$  no inconsistency is possible.

Portuguese notation in the tests:  $IC$  (= CI),  $RC$  (= CR),  $IA$  (= RI).

## When the judgements do not add up

Suppose that, in the same meeting, someone insists that payload is **twice as important** as cost ( $a_{13} = 1/2$ ), keeping the other five judgements.

Result:  $\lambda_{\max} = 4.467$  and  **$CR = 0.173 > 0.10$** . The weights become 0.343, 0.292, 0.276, 0.089 — but they cannot be trusted.

**Where is the problem?** Compare each judgement  $a_{ij}$  with what the weights imply,  $w_i/w_j$ : the ratio  $a_{ij} w_j/w_i$  should be close to 1.

$a_{ij} w_j/w_i$  with  $a_{13} = 1/2$

|         | cost        | range | payl.       | serv. |
|---------|-------------|-------|-------------|-------|
| cost    | 1           | 1.70  | <b>0.40</b> | 1.30  |
| range   | 0.59        | 1     | 1.89        | 0.92  |
| payload | <b>2.49</b> | 0.53  | 1           | 0.65  |
| service | 0.77        | 1.09  | 1.54        | 1     |

The cost–payload pair is the furthest off: that is the judgement to send back to the decision maker. With  $a_{13} = 3$  we are back to  $CR = 0.005$ ; with  $a_{13} = 1$  it is still acceptable ( $CR = 0.069$ ).

The AHP does not correct the decision maker: it shows where they contradict themselves.

## Two ways of obtaining the weights

### Column normalisation (used in class)

Average of the rows of the normalised matrix.

Easy by hand; it is the method of the tests and of the Excel sheet on Moodle.

### Eigenvector (Saaty)

$w$  is the eigenvector of  $A$  for the largest

eigenvalue,  $Aw = \lambda_{\max} w$ , normalised to sum to

1. This is the original definition; in Python, `numpy.linalg.eig`.

|                  | columns | eigenvector |
|------------------|---------|-------------|
| cost             | 0.4824  | 0.4829      |
| range            | 0.2718  | 0.2720      |
| payload          | 0.1575  | 0.1570      |
| service          | 0.0883  | 0.0882      |
| $\lambda_{\max}$ | 4.0145  | 4.0145      |

With consistent judgements, the two methods coincide; with  $CR < 0.10$  the differences appear in the third decimal place. When they differ a lot, the problem is the matrix, not the method.

## A qualitative criterion: after-sales service

For service there are no numbers: the company talked to other customers and compared the brands in pairs.

| service | A   | B   | C | priority     |
|---------|-----|-----|---|--------------|
| A       | 1   | 1/3 | 3 | 0.261        |
| B       | 3   | 1   | 5 | <b>0.633</b> |
| C       | 1/3 | 1/5 | 1 | 0.106        |

$\lambda_{\max} = 3.039$ ;  $CI = 0.019$ ;  $CR = 0.033$ : consistent.

The same procedure works at **every** level:

1. compare the criteria  $\Rightarrow$  weights  $w$ ;
2. within each criterion, compare the alternatives  $\Rightarrow$  local priorities;
3. final score = weighted sum of the local priorities.

For cost, range and payload there are measured values: no need to ask for opinions. How to use them is the topic of **8.2**.

```
import numpy as np
IA = {3: 0.58, 4: 0.90, 5: 1.12, 6: 1.24, 7: 1.32}    # random index RI

def prioridades(A, metodo='colunas'):                # priorities (weights)
    A = np.asarray(A, float); n = len(A)
    if metodo == 'colunas':                          # column normalisation
        w = (A / A.sum(0)).mean(1)
        lam = np.mean(A @ w / w)
    else:                                             # eigenvector
        val, vec = np.linalg.eig(A); k = val.real.argmax()
        w = abs(vec[:, k].real); w /= w.sum(); lam = val[k].real
    IC = (lam - n) / (n - 1)
    return w, lam, IC / IA[n]                       # weights, lambda_max, CR

A = [[1, 2, 3, 5], [1/2, 1, 2, 3], [1/3, 1/2, 1, 2], [1/5, 1/3, 1/2, 1]]
print(prioridades(A))
# weights 0.482 0.272 0.158 0.088; lambda_max 4.015; CR 0.005
```

Full file: code/python/08\_ahp/  
08\_1\_ahp/ahp.py (function names in  
Portuguese, as in the course code).

**Try it:** change  $a_{13}$  to  $1/2$  and watch CR  
rise to 0.173. Which other judgement  
would have to change to make the  
matrix consistent again?

### Example 1 — comparing three fruits

A consumer compared three fruits, F1, F2 and F3, in pairs: F2 is moderately preferred to F1 (3), F1 is strongly preferred to F3 (5) and F2 is very strongly preferred to F3 (7). Find the priorities and check the consistency of the judgements.

### Example 2 — choosing between two job offers

A recent graduate has two offers, A and B, and three criteria. Criteria comparison matrix:

$\begin{pmatrix} 1 & 1/7 & 1/2 \\ 7 & 1 & 5 \\ 2 & 1/5 & 1 \end{pmatrix}$ . On criterion 1 (salary) A offers 2000 € and B 2500 €; on criterion 2, A is

moderately preferred to B (3); on criterion 3, A is strongly preferred to B (5). Which offer should they accept? Are the judgements consistent?

## Exercise sheet 8 — Exercise 1 (homework)

### AHP with 3 criteria and 3 alternatives (Python solution on Moodle)

We want to choose the best of three alternatives, A, B and C, using three criteria, C1, C2 and C3. Criteria

matrix:  $\begin{pmatrix} 1 & 1/2 & 4 \\ 2 & 1 & 7 \\ 1/4 & 1/7 & 1 \end{pmatrix}$ . Alternatives matrices for C1, C2 and C3:

$$\begin{pmatrix} 1 & 1/3 & 3 \\ 3 & 1 & 5 \\ 1/3 & 1/5 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 5 & 1/7 \\ 1/5 & 1 & 1/8 \\ 7 & 8 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 & 1/3 \\ 1/2 & 1 & 1/4 \\ 3 & 4 & 1 \end{pmatrix}.$$

Compute the weights, the consistency ratios of all matrices and the final score. Which alternative do you choose? Comment on the consistency.

Submission: by email to the lecturer, by the deadline given in class. All exercises: [Exercise sheet 8](#).

## KEY TAKEAWAYS — 8.1

1. The AHP is used when the criteria have no common unit: hierarchy goal  $\rightarrow$  criteria  $\rightarrow$  alternatives.
2. Saaty's scale from 1 to 9, reciprocal matrix,  $n(n-1)/2$  judgements.
3. Weights: normalise the columns and average the rows (or the eigenvector of  $\lambda_{\max}$ ).
4. Consistency:  $CI = (\lambda_{\max} - n)/(n - 1)$ ,  $CR = CI/RI_n < 0.10$ . If it fails, look for the judgement whose  $a_{ij}w_j/w_i$  is furthest from 1 and send it back to the decision maker.

Next: 8.2 Quantitative attributes and aggregation.

# References and resources

## References Full list: *Course bibliography*.

- Hillier, F. S., Lieberman, G. J., *Introduction to Operations Research*, 11th ed., McGraw-Hill, 2021 — Decision Analysis.
- Saaty, T. L., How to make a decision: the analytic hierarchy process, *European Journal of Operational Research*, 48(1), 9–26, 1990.
- Saaty, T. L., Decision making with the analytic hierarchy process, *International Journal of Services Sciences*, 1(1), 83–98, 2008.
- Taha, H. A., *Operations Research: An Introduction*, 10th ed., Pearson, 2017 — AHP.

## Resources

- Exercise sheet 8. `code/python/08_ahp/08_1_ahp/ahp.py`. Moodle: solutions in Python and MATLAB.